

Analysis 2: HW1

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February 12, 2026

Problem 2. 1. We have $f(x) = e^{-1/x}1_{(0,\infty)}$. Now if we represent $g(x) = e^x$ and $h(x) = -\frac{1}{x}$ then we have $f(x) = g(h(x))$ and therefore $f'(x) = g'(h(x))h'(x)$, which is $f'(x) = e^{-1/x} \frac{1}{x^2} 1_{(0,\infty)}$. However, note we still need to show that it's differentiable at 0. We have $\lim_{h \rightarrow 0} f(x+h) - f(x)/h = \lim_{h \rightarrow 0} (e^{-1/h} - 0)/h$. Clearly as e^x exponentially increases compared to x , the exponential term dominates and the terms converge to 0. So we can now write,

$$f'(x) = e^{-1/x} \frac{1}{x^2} 1_{(0,\infty)}$$

which means when $x \in (0, \infty)$ the derivative is $e^{-1/x} \frac{1}{x^2}$ otherwise it's 0. Now note we can also represent this as $f'(x) = f(x)h'(x)$. Now we need to show that for any $n \in \mathbb{N}$ that $f^{(n)}$ exists and is differential. We will do this using induction. We claim that for each $n \in \mathbb{N}$, $f^{(n)}$ is of the form $f(x)P_n(h)$ where P_n is some polynomial representation of the derivatives of h from 1 to n . Note that h is smooth as we have $h^{(n)} = (-1)^{n+1} \frac{1}{x^{n+1}}$. If our inductive hypothesis is true this means that each of our derivatives i.e $f^{(n)}$ is the product of two differentiable functions and we can then easily show the product is also differentiable by giving a formulation for the derivative. Our base case is trivial as we just showed it above i.e $f'(x) = f(x)h'(x)$. Now assume true for case i , so we have,

$$f^{(i)}(x) = f(x)P_n(h)$$

now find the derivative of this and we get,

$$f^{(i+1)}(x) = f'(x)P_n(h) + f(x)P_n(h)'$$

But $f'(x) = f(x)h'(x)$ so,

$$\begin{aligned} f^{(i+1)}(x) &= f(x)h'(x)P_n(h) + f(x)P_n(h)' \\ &= f(x)(h'(x)P_n(h) + P_n(h)') \end{aligned}$$

Note however that $h'(x)P_n(h) + P_n(h)'$ will return just another polynomial say $P'_{n+1}(h)$ and hence is of the same form. So we have,

$$f^{(i+1)} = f(x)P'_{n+1}(h)$$

Hence we show by induction that this is true for any arbitrary n . Now note that differentiability at 0 is also guaranteed because of the exponential domination and is implicitly contained in the above formulation because of the definition of f (0 at 0).

2. Consider the following function,

$$H(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ \frac{f(x)}{f(1-x)+f(x)} & \text{for } 0 < x \leq 1 \\ 1 & \text{otherwise} \end{cases}$$

3. We essentially need ψ to be 1 inside $B(0,1)$, between 0 and 1 between the two balls and 0 outside the $B(0,2)$ ball. For a real valued function in R^2 we have,

$$\psi(x, y) = \begin{cases} 1 & \text{for } \sqrt{x^2 + y^2} \leq 1 \\ \frac{f(2-\sqrt{x^2+y^2})}{f(2-\sqrt{x^2+y^2})+f(\sqrt{x^2+y^2}-1)} & \text{for } 1 < \sqrt{x^2 + y^2} < 2 \\ 0 & \text{for } 2 \leq \sqrt{x^2 + y^2} \end{cases}$$

And for ψ in $\mathbb{R}^n$ consider $x = (x_1, \dots, x_n)$, and denote $r = |x| = \sqrt{x_1^2 + \dots + x_n^2}$. Then our function is,

$$\psi(x) = \begin{cases} 1 & \text{for } r \leq 1 \\ \frac{f(2-r)}{f(2-r)+f(r-1)} & \text{for } 1 < r < 2 \\ 0 & \text{for } 2 \leq r \end{cases}$$